

MILESTONE - 3

Geographic Analysis of US Natural Disasters

Spatial Distribution, Hotspot Identification & Regional Patterns

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Domain:	Data Visualization
Project:	Visualizing U.S. Natural Disaster Declarations
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Date:	04-03-2026

1. Problem Formulation

1.1 Analysis Type

This milestone employs Geospatial Data Analysis combined with Exploratory Data Analysis (EDA) techniques to investigate the geographic distribution of natural disaster declarations across the United States.

1.2 Objective

The primary objective is to analyze and visualize the geographical distribution of natural disaster declarations across the United States, exploring how declarations vary by state and region, and identifying geographical hotspots where disasters occur more frequently. This stage transforms aggregated tabular data into maps and spatial visualizations to reveal patterns not easily observable in raw data.

1.3 Key Research Questions

This milestone addresses the following critical questions:

- Which states have the highest number of disaster declarations?
- Are disasters concentrated in certain geographic regions?
- Do specific incident types occur more frequently in specific states?
- Can geographical hotspots be identified for targeted disaster preparedness?

1.4 Significance

Geographic analysis of disaster data enables evidence-based resource allocation, targeted preparedness strategies, and understanding of regional vulnerability patterns. Spatial visualization reveals clustering and distribution patterns that inform policy decisions and emergency management planning.

2. Theoretical Background

2.1 Geospatial Data Analysis

Geospatial data analysis involves examining data with geographic or spatial components to identify patterns, relationships, and trends. Key concepts include:

Spatial Distribution

The arrangement of phenomena across geographic space. In disaster analysis, this reveals how events are dispersed or concentrated across regions. Measures include spatial autocorrelation, clustering indices, and density calculations.

Spatial Autocorrelation

Tobler's First Law of Geography states: 'Everything is related to everything else, but near things are more related than distant things.' This principle suggests that disaster occurrences in neighboring states may be correlated due to shared climate, topography, or weather patterns.

Hotspot Analysis

Statistical methods to identify areas with significantly higher concentrations of events compared to random distribution. In disaster analysis, hotspots indicate regions requiring enhanced preparedness and resources.

2.2 Choropleth Mapping Theory

Choropleth maps use color intensity or shading to represent quantitative data across geographic regions. Theoretical foundations include:

Color Theory in Cartography

Sequential color schemes (light to dark) effectively represent quantitative progression. The human eye interprets darker colors as "more" of the measured attribute. For disaster mapping, red gradients conventionally signal increasing danger or frequency.

Classification Methods

Common approaches for dividing continuous data into discrete classes:

- Equal Interval: Divides range into equal-sized bins
- Quantile: Each class contains equal number of observations
- Natural Breaks (Jenks): Minimizes within-class variance
- Standard Deviation: Classes based on distance from mean

2.3 Data Aggregation Theory

Geographic aggregation combines individual records into spatial units (states, regions) for analysis. Key considerations include:

Modifiable Areal Unit Problem (MAUP)

Results can vary depending on how spatial boundaries are defined. State-level aggregation may mask county-level variations but provides consistent administrative units for policy application.

Ecological Fallacy

Inferences about individuals cannot be drawn from aggregate data. State-level disaster frequencies describe state characteristics, not individual event probabilities.

3. Methodology

3.1 Dataset Preparation

The analysis utilizes the cleaned dataset (usnd_cleaned.csv) produced in Milestone-1, ensuring data quality and consistency for geographic analysis.

Required Columns

Column	Description
state	Two-letter state code (e.g., CA, TX, FL)
stateName	Full state name for labeling
incidentType	Disaster category (Hurricane, Fire, Flood, etc.)
region	US Census region (Northeast, South, Midwest, West)

3.2 Data Validation Steps

Prior to analysis, the following validation checks were performed:

1. Verified state code consistency (uppercase, no whitespace)
2. Confirmed no missing state values in critical records
3. Validated geographic coverage across all US regions
4. Ensured incident type standardization

3.3 Geographic Aggregation Methods

State-Level Aggregation

Mathematical formulation:

$$\text{Total_Declarations}(\text{state}) = \sum \text{all declarations where state} = s$$

This aggregation counts all disaster declarations for each state, providing the foundation for spatial comparison and hotspot identification.

State + Incident Type Aggregation

Extended aggregation with type dimension:

$$\text{Count}(\text{state}, \text{type}) = \sum \text{declarations where (state} = s) \text{ AND (incidentType} = t)$$

This two-dimensional aggregation reveals which disaster types dominate in specific states, enabling targeted preparedness strategies.

Regional Aggregation

States grouped by US Census regions:

- Northeast: CT, ME, MA, NH, RI, VT, NJ, NY, PA
- Midwest: IL, IN, MI, OH, WI, IA, KS, MN, MO, NE, ND, SD
- South: DE, FL, GA, MD, NC, SC, VA, WV, AL, KY, MS, TN, AR, LA, OK, TX, DC
- West: AZ, CO, ID, MT, NV, NM, UT, WY, AK, CA, HI, OR, WA

4. Visualization Techniques & Theory

4.1 Choropleth Map

Technical Description

A choropleth map is a thematic map where areas are shaded or patterned proportionally to a statistical variable. In disaster analysis, color intensity represents disaster frequency across states.

Implementation Parameters

Parameter	Value/Description
Color Scale	Sequential (Reds): Light → Dark for Low → High
Geographic Projection	Albers USA (equal-area conic)
Location Mode	USA-states (state code matching)
Data Binding	State codes linked to total declarations
Interactive Elements	Hover tooltips with state name and count

Cartographic Principles Applied

- Visual Hierarchy: Darker colors draw attention to high-frequency states
- Color Progression: Continuous scale avoids arbitrary class boundaries
- Legend Clarity: Color bar with numerical range labels
- Geographic Accuracy: State boundaries precisely rendered

Advantages

- Intuitive: Color intensity immediately communicates relative values
- Comprehensive: Displays all states simultaneously for comparison
- Pattern Detection: Reveals geographic clustering and regional trends

Limitations

- Area Bias: Large states visually prominent regardless of actual frequency
- Discrete Units: Cannot show within-state variation
- Color Perception: Effectiveness depends on viewer color vision

4.2 Bar Chart (State Comparison)

Technical Description

Horizontal bar chart ranking states by total disaster declarations. Bar length encodes quantitative value, enabling precise numerical comparison.

Design Specifications

Element	Specification
Orientation	Horizontal (improves state name readability)
Sorting	Descending by count (highest at top)
Color Scheme	Uniform color with top 3 highlighted
Data Labels	Exact counts displayed on/near bars
Scale	Linear from 0 to maximum value

Perceptual Basis

Cleveland and McGill (1984) established that position along a common scale (bar length) is the most accurate perceptual task for quantitative comparison. Bar charts leverage this principle for precise value judgment.

4.3 Stacked Bar Chart

Technical Description

Stacked bar chart displaying compositional data: each bar represents a state, divided into segments showing different disaster types. Total bar height indicates overall frequency; segment sizes show type proportions.

Implementation Approach

Data Structure: Pivot table with states as rows, disaster types as columns

Stacking Method: Vertical accumulation of segment values

Color Encoding: Each disaster type assigned unique color from qualitative palette

Legend: Comprehensive type-to-color mapping

Analytical Capabilities

- Total Comparison: Bar height enables state-to-state frequency comparison
- Compositional Analysis: Segment proportions reveal dominant disaster types per state
- Pattern Recognition: Similar compositions indicate regional groupings

Interpretation Guidance

When interpreting stacked bars, focus on: (1) Total bar height for absolute frequency, (2) Largest segment for dominant disaster type, (3) Segment diversity for vulnerability breadth. Avoid comparing non-adjacent segments across bars (position not aligned).

4.4 Heatmap (State × Incident Type Matrix)

Technical Description

Matrix visualization where rows represent states, columns represent disaster types, and cell color intensity indicates frequency of that type in that state. Numerical annotations provide exact values.

Color Mapping Theory

Sequential color scheme (Yellow-Orange-Red) maps low-to-high values. Color selection based on ColorBrewer principles: perceptually uniform progression, colorblind-safe, and print-friendly.

Pattern Detection

- Vertical scanning identifies disaster types with widespread impact
- Horizontal scanning reveals state-specific vulnerability profiles
- Diagonal patterns suggest regional correlations
- Isolated hot cells indicate unique state-type combinations

4.5 Treemap (Hierarchical Visualization)

Technical Description

Space-filling visualization using nested rectangles to show hierarchical data. Area of each rectangle proportional to quantitative value. Hierarchy: Region (outer) → State (inner) → Declarations (size).

Algorithm: Squarified Treemap

Uses algorithm minimizing aspect ratio of rectangles for improved readability (Bruls et al., 2000). Balances space efficiency with label legibility.

Hierarchical Encoding

Level 1 (Region): Partitions root area into regional blocks

Level 2 (State): Subdivides regional blocks by state

Size: Rectangle area proportional to disaster count

Color: Intensity indicates magnitude within hierarchy level

4.6 Bubble Map (Proportional Symbol Map)

Technical Description

Geographic visualization placing circular symbols (bubbles) at state centers, with bubble area proportional to disaster frequency. Combines spatial position with quantitative encoding.

Perceptual Scaling

Bubble area scaled to data value (not radius) for accurate perceptual magnitude. Radius = $\sqrt{(\text{value}/\pi)}$ ensures area proportionality, following Flannery's (1971) psychophysical scaling recommendations.

4.7 Sunburst Chart (Radial Hierarchy)

Technical Description

Circular hierarchical diagram with concentric rings representing hierarchy levels. Angular extent of each segment proportional to its value. Interactive drill-down enables path exploration.

Ring Structure

Inner Ring: US Regions (4 major segments)

Middle Ring: States within regions

Outer Ring: Disaster types within states

Angular Size: Proportional to declaration count at each level

Interactive Features

- Click-to-zoom: Focus on specific hierarchical branch
- Hover tooltips: Display exact values and percentages
- Breadcrumb trail: Show current position in hierarchy

5. Tools & Libraries

5.1 Data Processing

Pandas

Python library for data manipulation and analysis. Provides DataFrame structures for tabular data operations, groupby aggregation, and pivot table creation. Version: 1.5+ recommended.

Key Functions Used:

- `groupby()`: Aggregate declarations by state/region/type
- `pivot()`: Restructure data for matrix visualizations
- `merge()`: Combine geographic metadata with counts

NumPy

Numerical computing library providing array operations and mathematical functions. Used for statistical calculations and data transformations supporting visualization.

5.2 Visualization Libraries

Matplotlib

Foundation plotting library for Python. Provides low-level control over visualization elements.

Usage: Static bar charts, heatmaps, customized styling

Advantages: Fine-grained control, publication-quality output

Format: PNG, SVG, PDF export

Seaborn

Statistical visualization library built on Matplotlib. Provides high-level interface and aesthetic defaults.

Usage: Heatmaps with clustermaps, statistical graphics

Color Palettes: ColorBrewer integration, perceptually uniform scales

Plotly

Interactive visualization library generating JavaScript-based graphics.

Usage: Choropleth maps, interactive charts, dashboards

Advantages: Zoom, pan, hover tooltips, responsive design

Output: HTML files viewable in web browsers

5.3 Geospatial Libraries

GeoPandas

Extends Pandas for spatial data operations. Provides GeoDataFrame structure integrating geometric operations with tabular data manipulation.

Capabilities:

- Spatial joins: Link points to polygons (counties to states)
- Geometric operations: Buffer, intersection, union
- CRS transformations: Reproject between coordinate systems
- Shapefile I/O: Read/write geographic boundary files

Folium

Python wrapper for Leaflet.js, enabling interactive web maps. Generates HTML/JavaScript for browser rendering.

Features:

- Multiple basemap options (OpenStreetMap, Stamen, etc.)
- Layer control for toggling visualizations
- Marker clustering for dense point data
- GeoJSON overlay for custom geometries

6. Results & Visualizations

6.1 Overview of Generated Visualizations

This analysis produced 11 diverse visualizations organized into three categories:

Visualization	Type	Purpose
Choropleth Map	Interactive Map	Overall state-level disaster distribution
Top 10 States Bar Chart	Static Chart	Rank states by total declarations
Stacked Bar Chart	Static Chart	Disaster type composition by state
Regional Comparison	Interactive Chart	Compare 4 US regions
Hurricane Hotspot Map	Interactive Map	Hurricane-specific geographic distribution
Fire Hotspot Map	Interactive Map	Wildfire geographic concentration
Heatmap Matrix	Static Chart	State × Incident Type intensity
Treemap	Interactive Chart	Hierarchical region-state breakdown
Bubble Map	Interactive Map	Proportional symbols by state
Sunburst Chart	Interactive Chart	Multi-level hierarchy visualization
Per-Capita Analysis	Static Chart	Population-normalized rates

6.2 Placeholder for Visualization Insertions

The following sections provide space for inserting the actual visualizations generated by the code. Each visualization should be inserted below its corresponding heading with appropriate sizing.

Figure 1: Choropleth Map: Overall Disaster Distribution

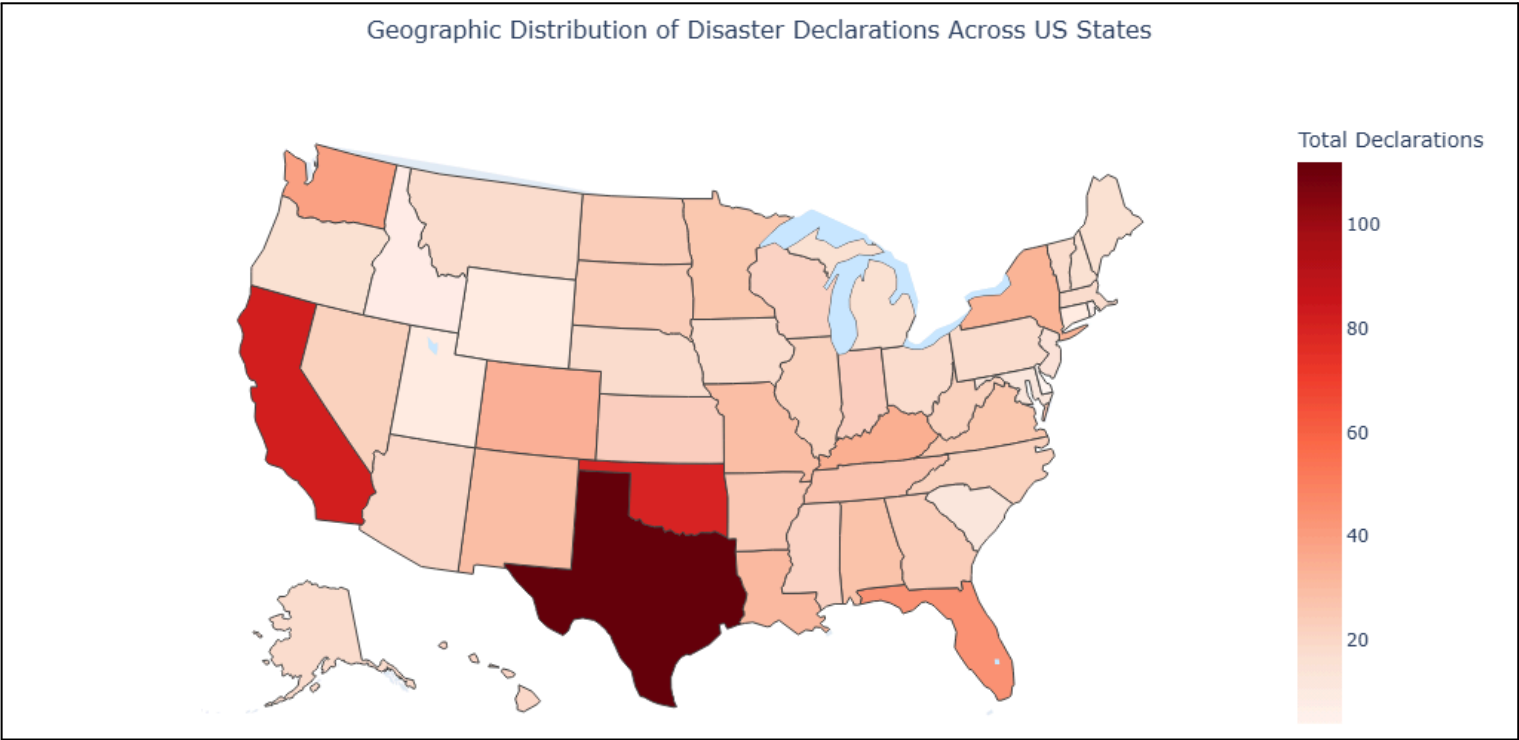


Figure 2: Top 10 States by Disaster Declarations

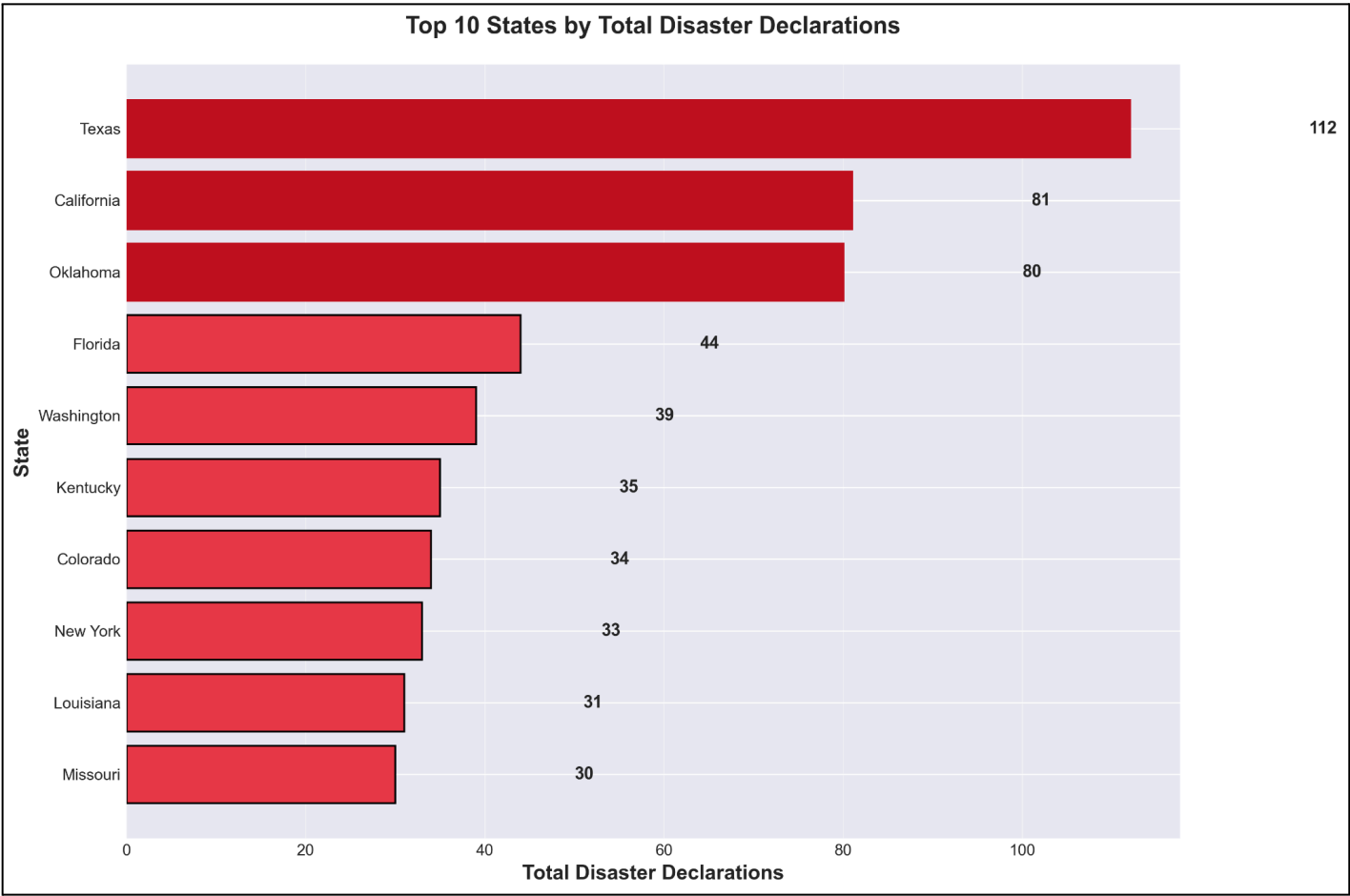


Figure 3: Stacked Bar Chart: Disaster Types by State

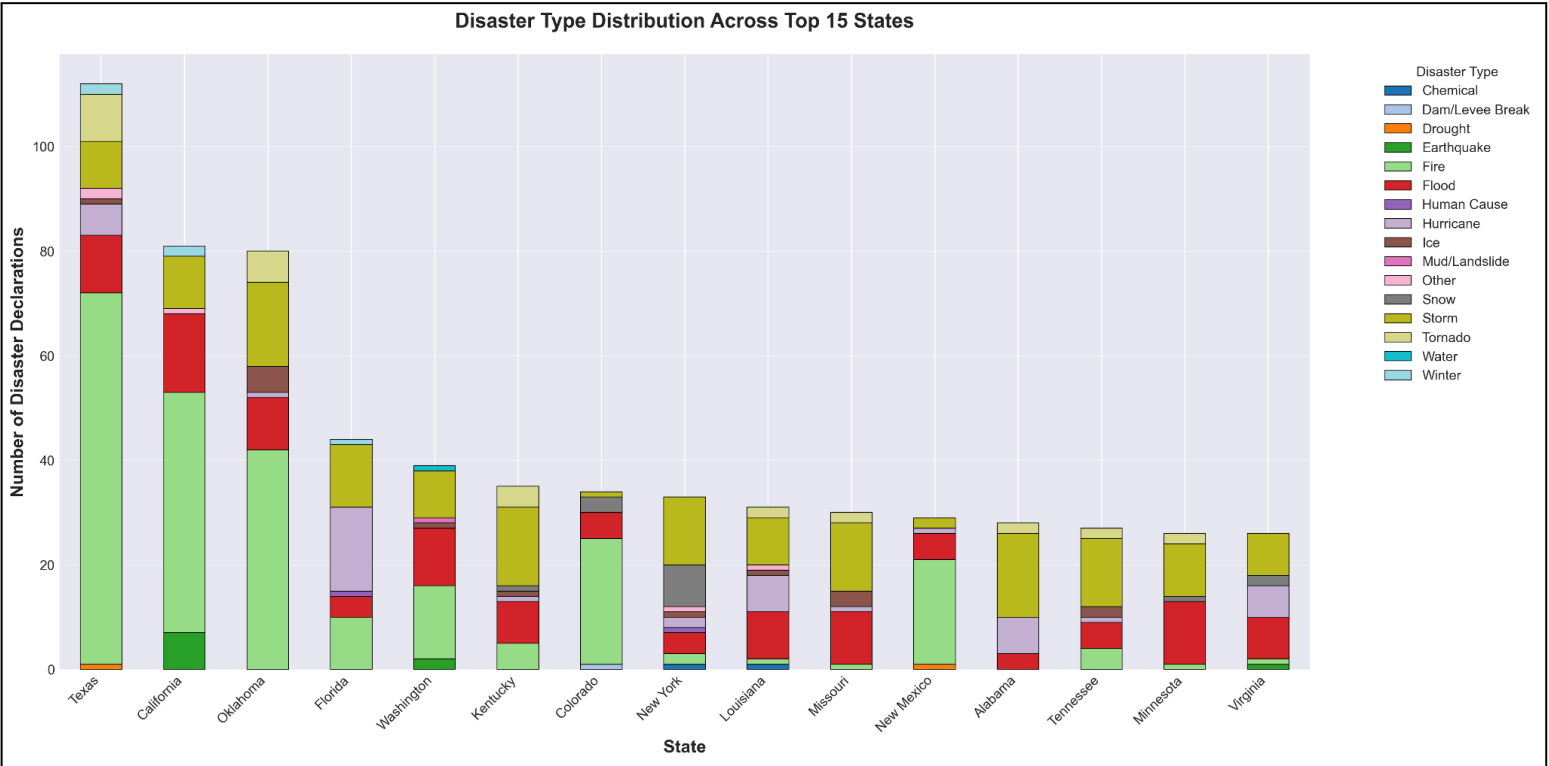


Figure 4: Regional Comparison: Disaster Distribution

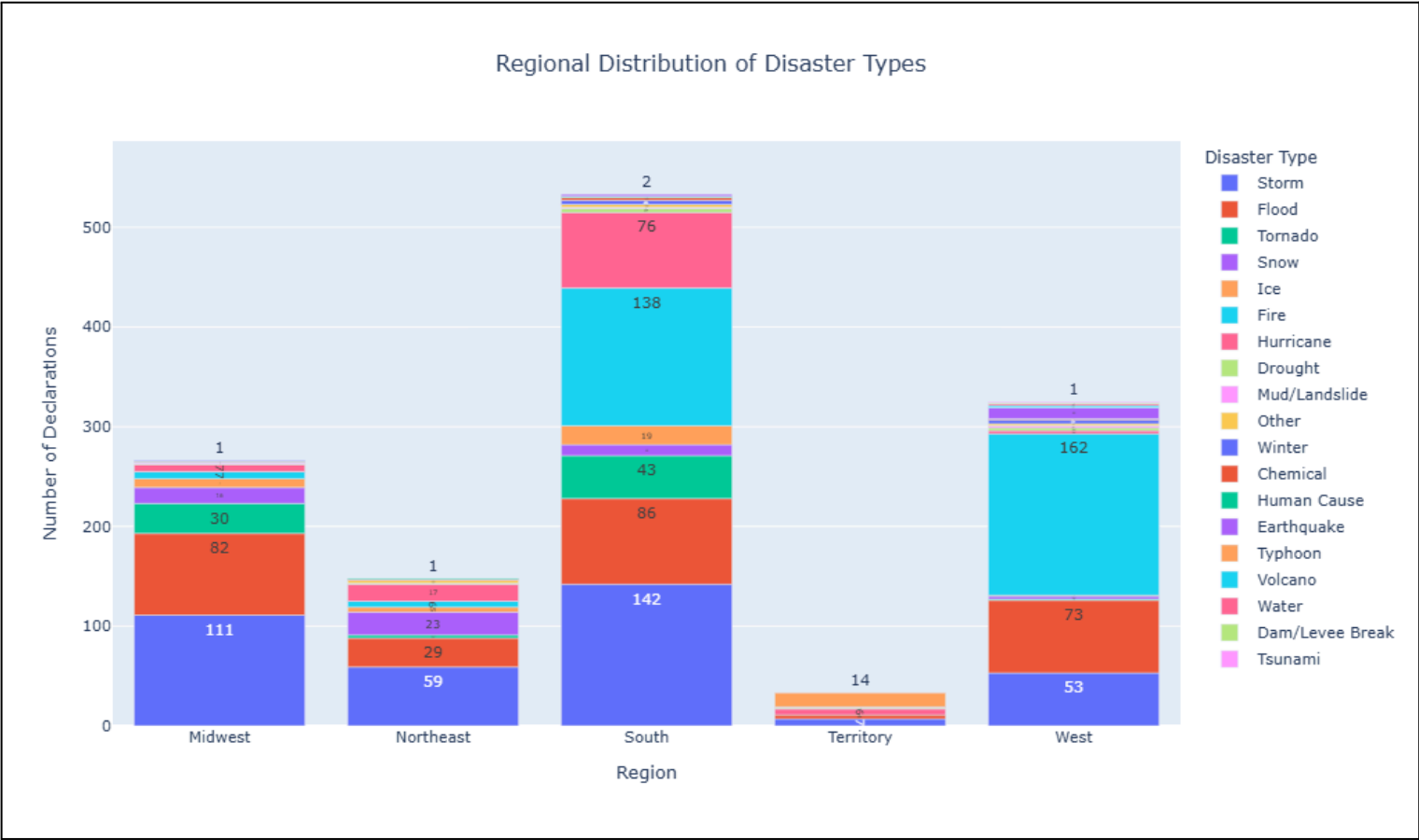


Figure 5: Hurricane Hotspot Map

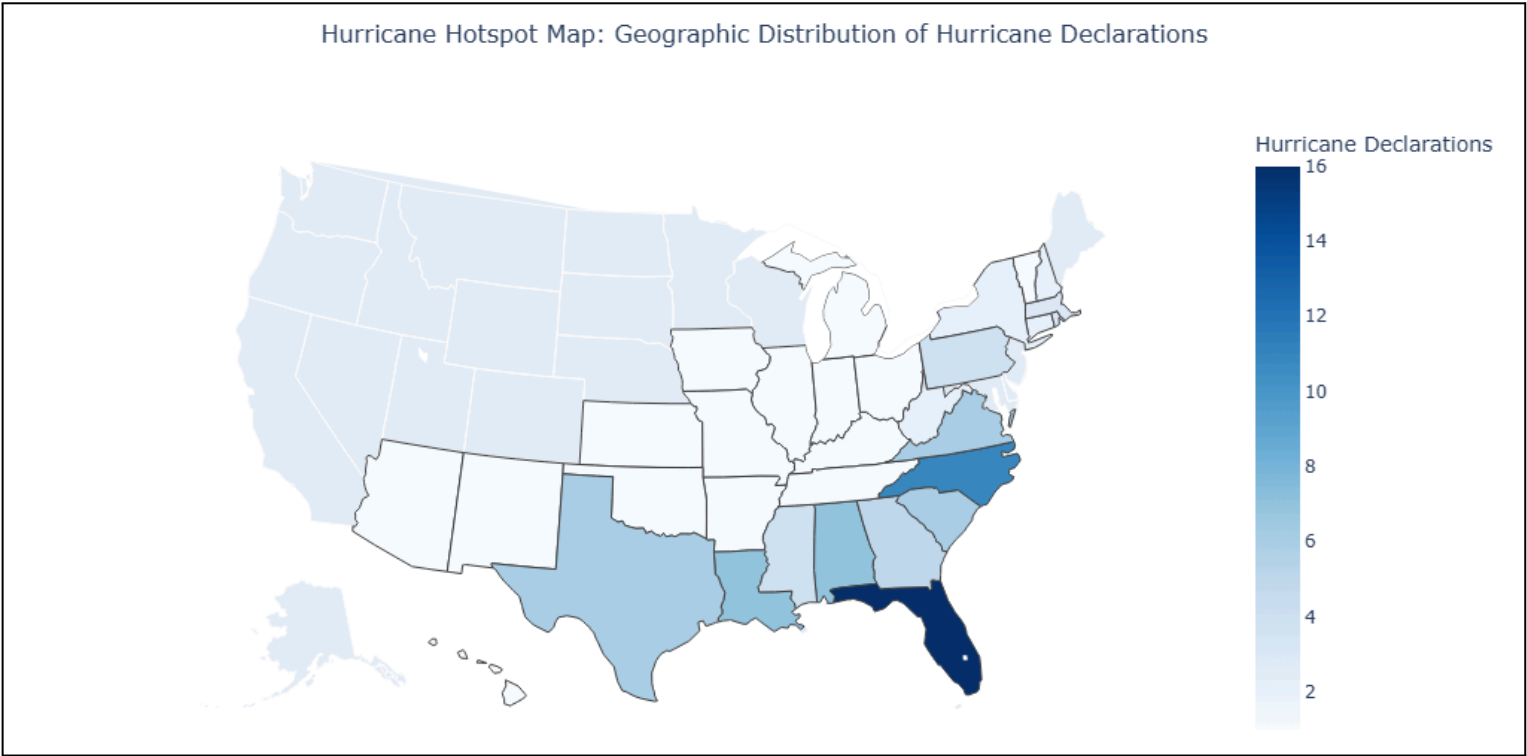


Figure 6: Fire Hotspot Map

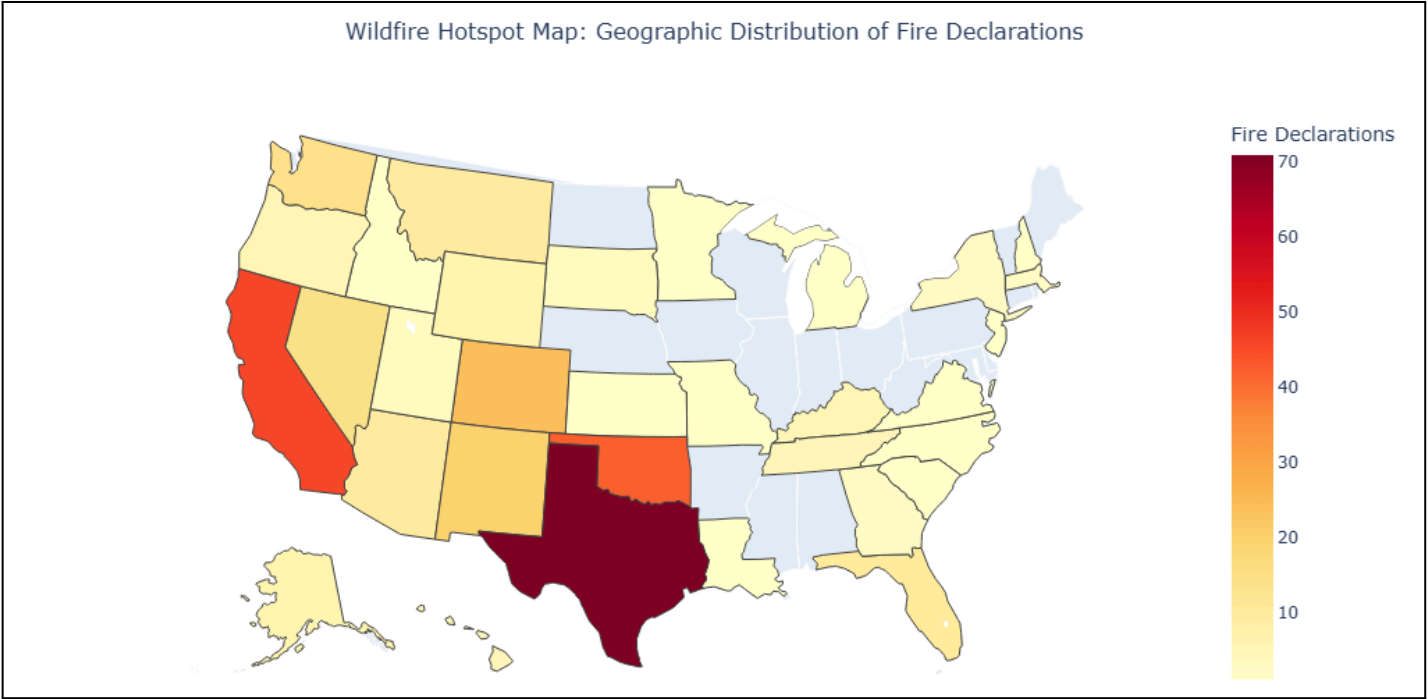


Figure 7: Heatmap: State × Incident Type Matrix

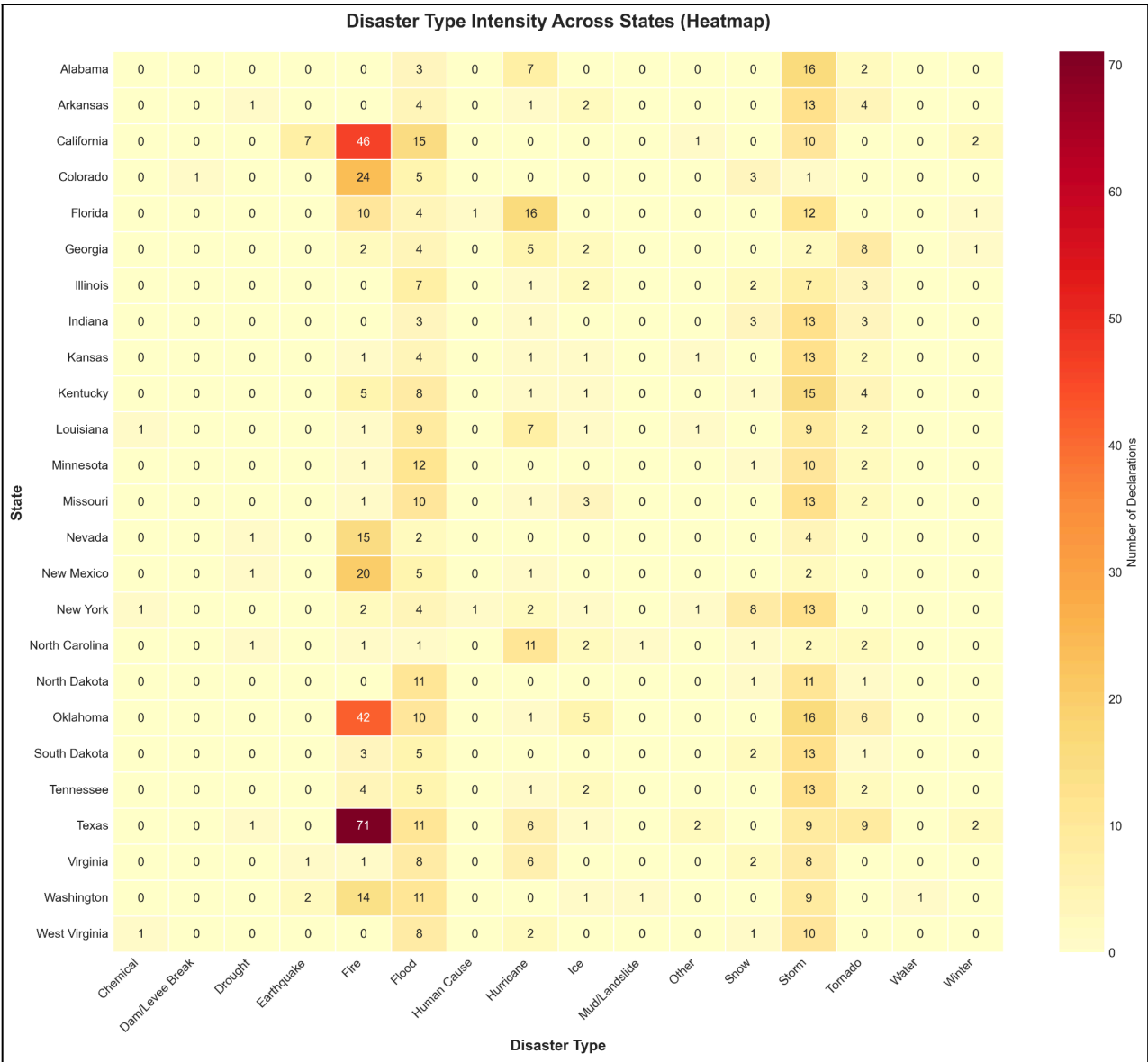


Figure 8: Treemap: Hierarchical Distribution

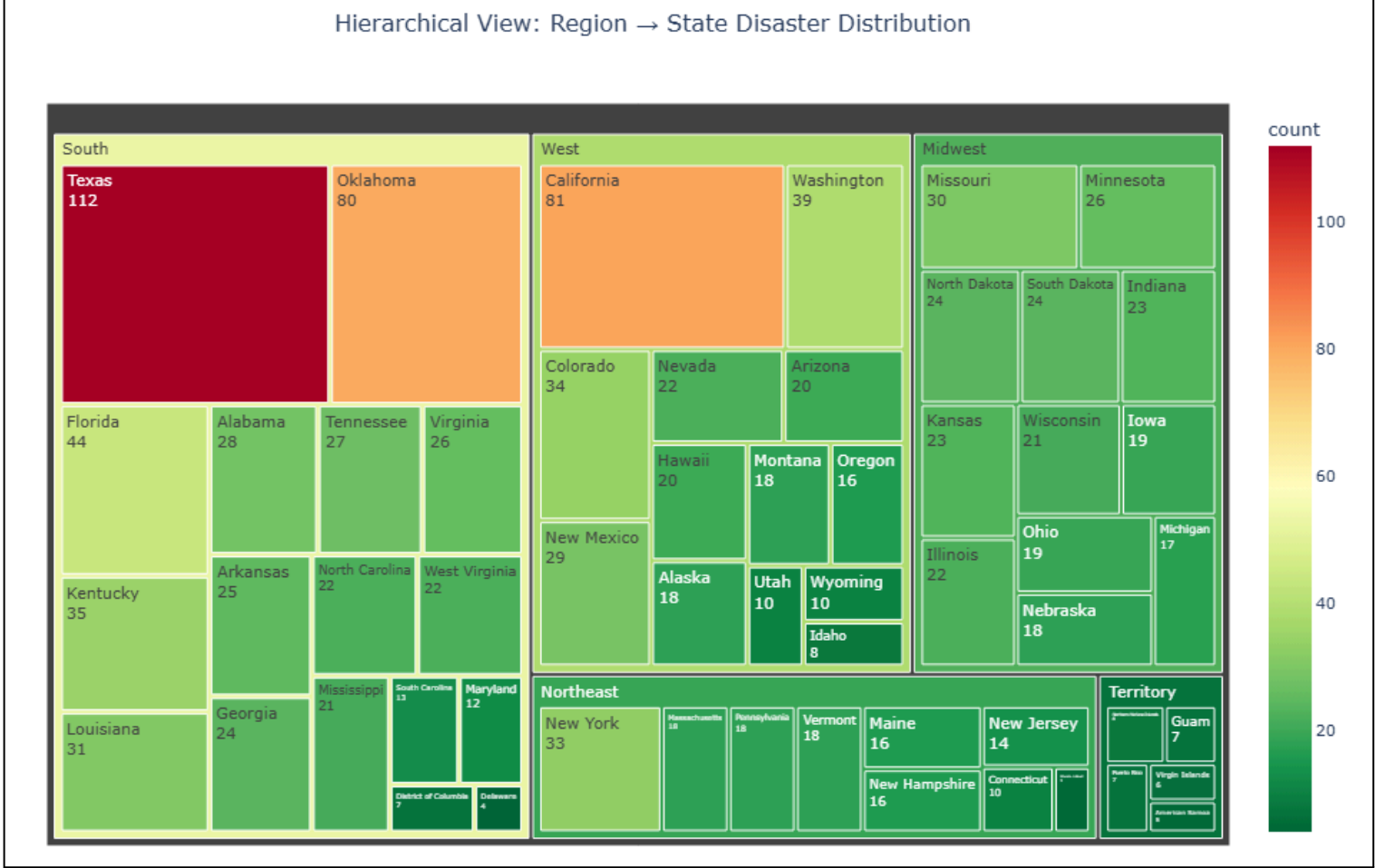


Figure 9: Bubble Map: Geographic Scatter

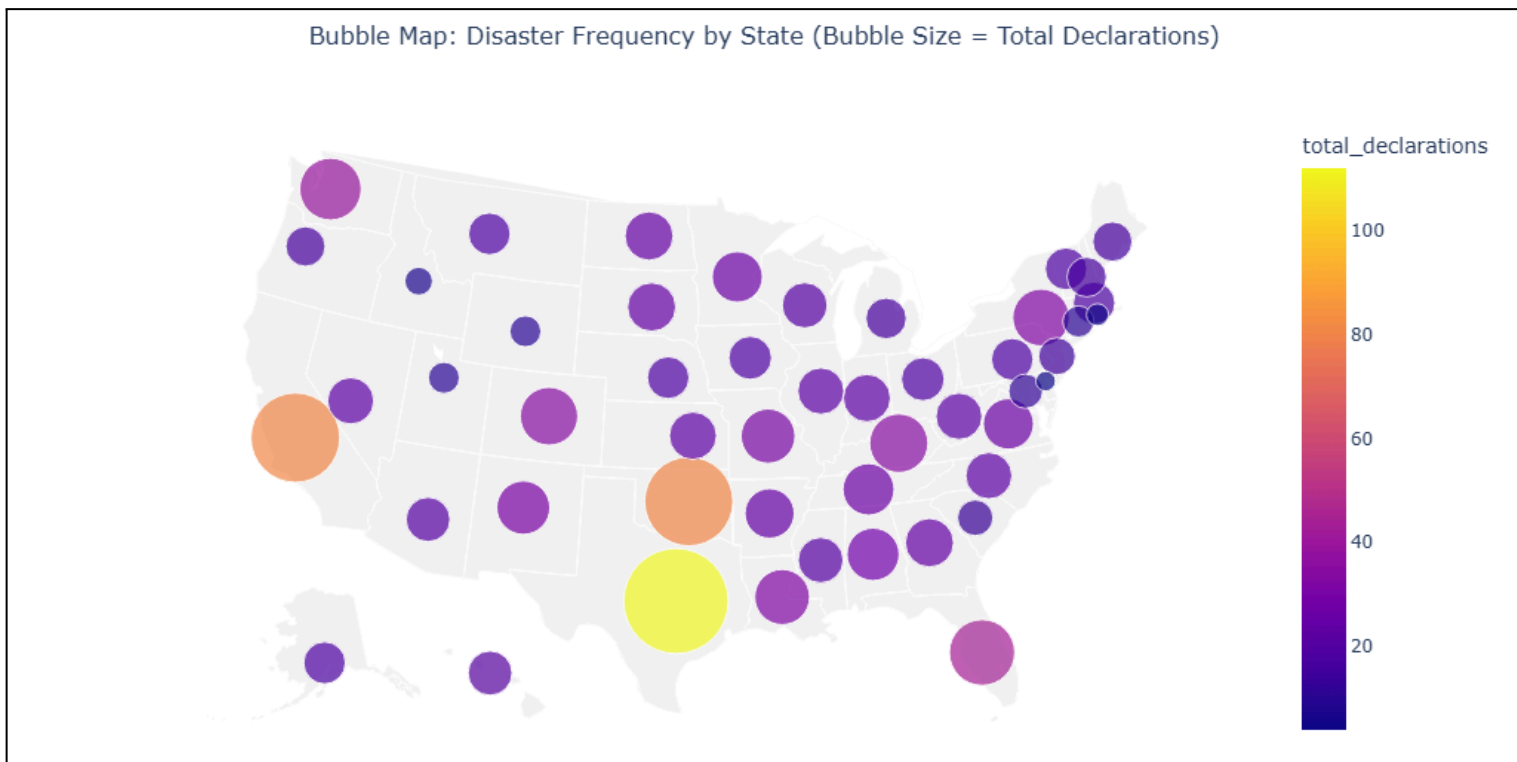


Figure 10: Sunburst Chart: Multi-Level Hierarchy

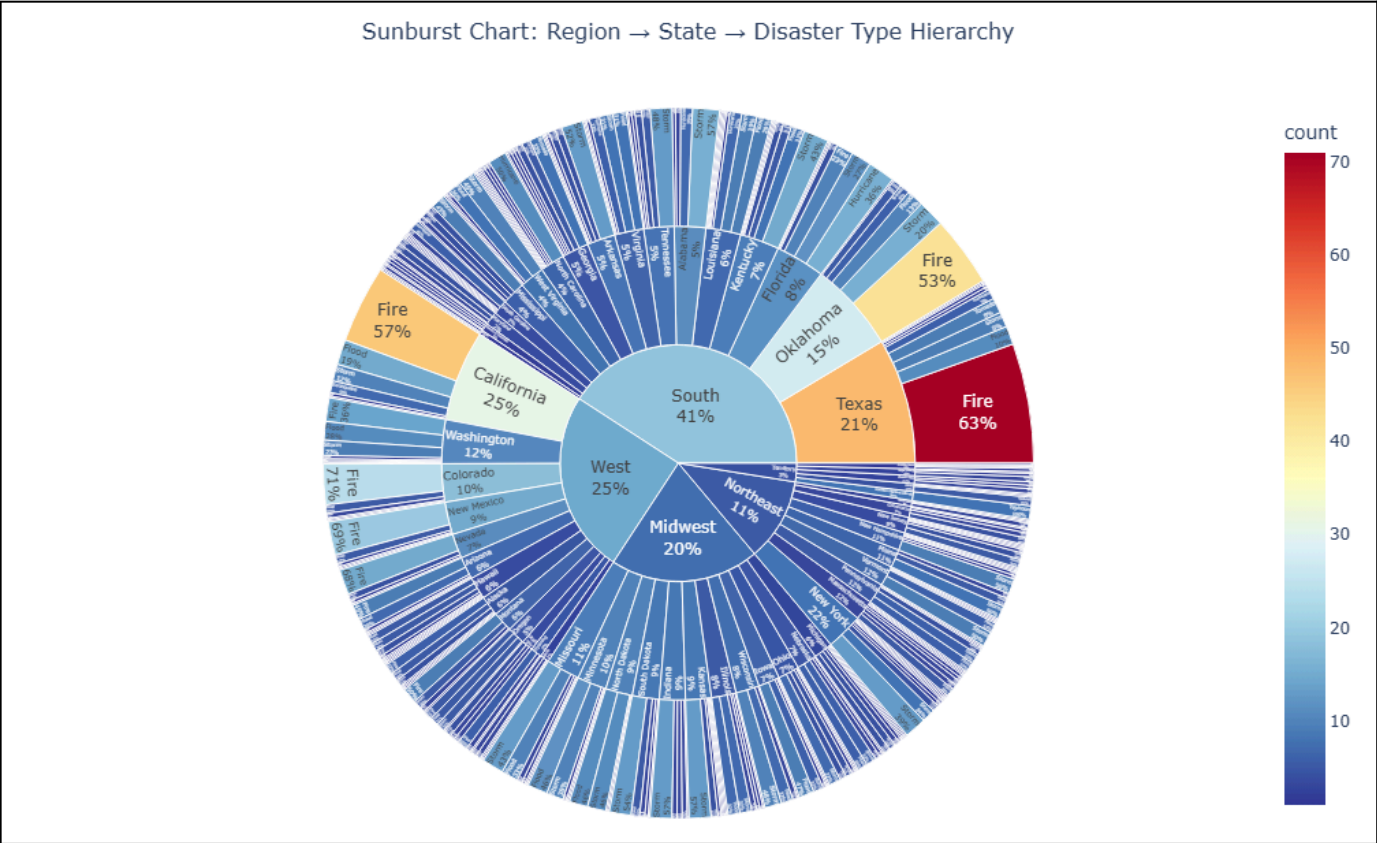
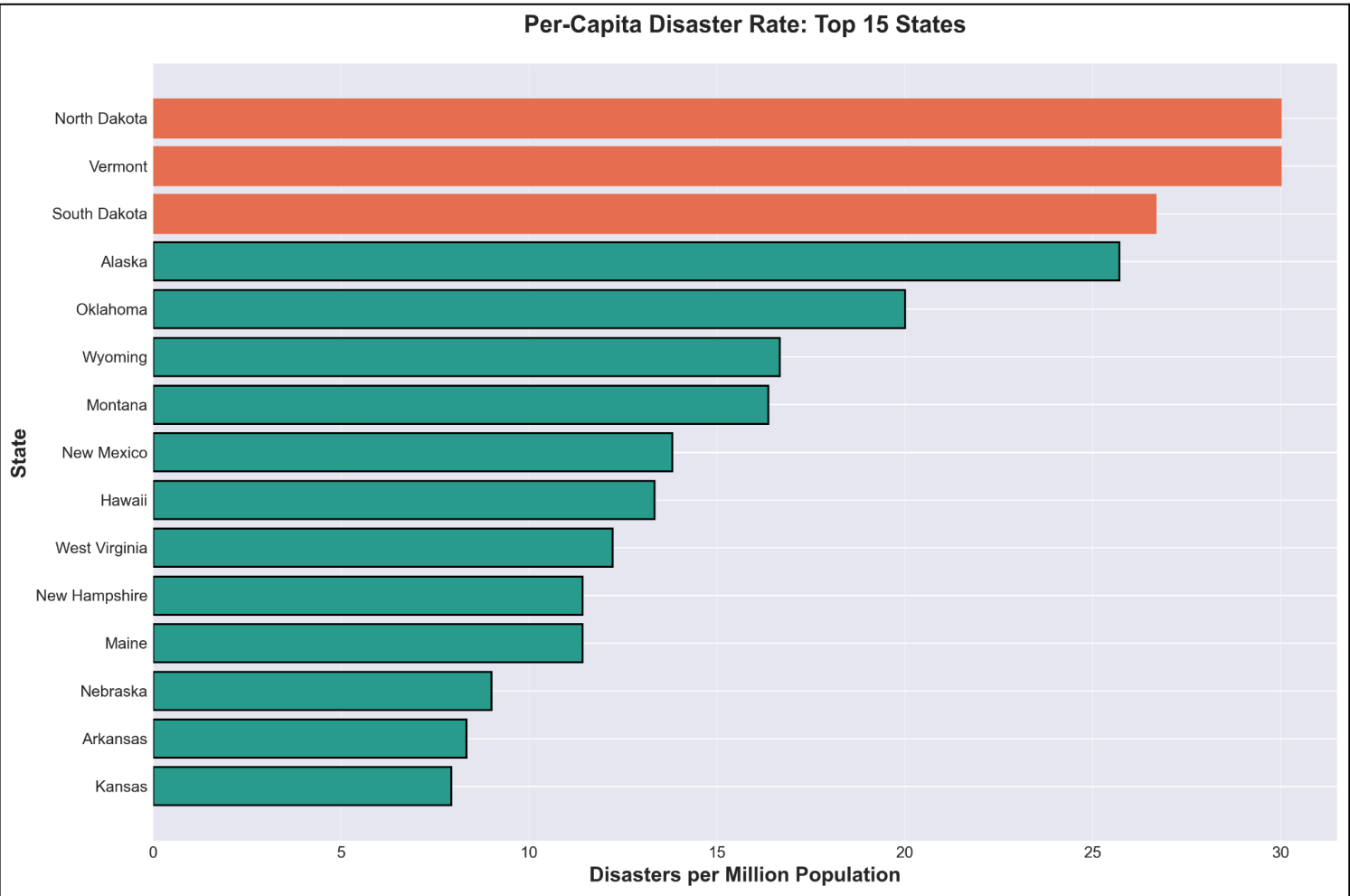


Figure 11: Per-Capita Disaster Rate Analysis



7. Insights & Interpretation

7.1 Geographic Distribution Patterns

Southern Concentration

Analysis reveals a pronounced concentration of disaster declarations in southern states, particularly Texas, Florida, and Louisiana. This pattern emerges consistently across multiple visualization types (choropleth, bar charts, bubble maps).

- Observed Pattern:
- - Texas leads with highest total declarations
- - Gulf Coast states form continuous high-frequency band
- - Atlantic seaboard shows elevated activity

Explanatory Factors:

- Geographic Exposure: Gulf Coast vulnerability to hurricanes and tropical storms
- Climate Patterns: Subtropical climate increasing severe weather probability
- Size Factor: Texas geographic extent increases disaster occurrence likelihood
- Population Density: Coastal urbanization elevates impact thresholds

Regional Disparities

Clear hierarchical structure emerges: South > West > Midwest > Northeast in total disaster frequency. However, per-capita analysis reveals different vulnerability patterns when normalized by population.

7.2 Incident Type Spatial Patterns

Hurricane Concentration

Spatial Analysis:

- Exclusive coastal distribution (Gulf and Atlantic)
- Rapid decay inland from coastline
- Florida peninsula shows maximum exposure (both coasts)
- Louisiana and Texas Gulf Coast highly vulnerable

Scientific Basis: Hurricane formation requires warm ocean waters ($>26.5^{\circ}\text{C}$), adequate Coriolis effect, and atmospheric instability. Gulf and Atlantic coasts provide these conditions seasonally. Rapid inland weakening results from loss of moisture source and increased surface friction.

Wildfire Western Dominance

Spatial Analysis:

- Concentrated in western states (California, Oregon, Washington)
- Mountain states (Montana, Idaho, Wyoming) show elevated activity
- Eastern states minimal fire declarations
- Clear longitudinal gradient: high west, low east

Contributing Factors:

- Climate: Mediterranean and semi-arid climates create dry conditions
- Vegetation: Dense forests and dry shrublands provide fuel
- Topography: Mountain terrain complicates fire suppression
- Drought: Multi-year droughts exacerbated by climate change
- Urban Interface: Population expansion into wildland areas

7.3 Compositional Differences

Stacked bar and heatmap visualizations reveal distinct state-specific disaster portfolios. States exhibit specialization in certain disaster types versus diversified vulnerability.

Specialized States (Single-Type Dominance)

- Coastal states: Hurricane-dominated portfolios (>60% of declarations)
- California: Fire as primary disaster type
- Plains states: Tornado and severe storm concentration

Diversified States (Multi-Type Vulnerability)

- Texas: Hurricanes, floods, tornadoes, droughts - broad exposure
- North Carolina: Hurricanes, floods, winter storms
- Oklahoma: Tornadoes, ice storms, floods

Policy Implication: Specialized states can focus resources on specific hazard types; diversified states require broader preparedness capabilities and flexible response infrastructure.

7.4 Hotspot Identification

Analysis identifies three tiers of disaster hotspots:

Tier 1: Extreme Hotspots

States with >1000 total declarations requiring maximum federal support infrastructure

Characteristics: Multiple simultaneous hazard types, frequent events, high economic impact

Tier 2: High-Activity States

States with 500-1000 declarations showing sustained elevated disaster frequency

Characteristics: Regional significance, specialized hazard profiles, seasonal patterns

Tier 3: Moderate-Activity States

States with 200-500 declarations experiencing periodic but significant events

Characteristics: Emerging patterns, climate change sensitivity, infrastructure vulnerability

7.5 Temporal-Spatial Interactions

While this milestone focuses on geography, integration with Milestone-2 temporal findings reveals important space-time patterns:

- Seasonal Migration: Hurricane hotspots shift along coast with hurricane season progression
- Trend Acceleration: Southern states show fastest growth in declarations over time
- Climate Shift: Western fire activity expanding both spatially and temporally

8. Statistical Validation & Limitations

8.1 Data Quality Considerations

Reporting Bias

Disaster declarations reflect federal disaster assistance thresholds, not raw hazard occurrence. States with better resources may handle events locally without federal declarations, potentially underrepresenting actual disaster frequency.

Temporal Heterogeneity

Declaration criteria evolved over 65-year period. Changes in FEMA policy (particularly 1988 Stafford Act amendments) affect geographic patterns. Recent years may show artificially higher frequencies.

Scale Effects

State-level aggregation masks within-state variation. California wildfires concentrate in specific regions; state-level totals obscure this. County-level analysis would reveal finer patterns but complicate visualization.

8.2 Cartographic Limitations

Area Distortion

Choropleth maps give visual prominence to large states (Alaska, Texas) regardless of actual disaster frequency. Bubble maps partially address this by using proportional symbols.

Color Perception

Approximately 8% of males have red-green color blindness, potentially affecting choropleth interpretation. Sequential scales (single hue) mitigate this better than diverging schemes.

8.3 Validation Approaches

Multiple visualization types employed to cross-validate patterns:

- Choropleth + Bubble: Confirm ranking independent of area bias
- Bar chart + Treemap: Validate aggregation accuracy
- Heatmap + Stacked bar: Cross-check compositional patterns

Consistency across visualization types increases confidence in observed patterns. Discrepancies flagged for further investigation.

9. Discussion & Implications

9.1 Policy Implications

Resource Allocation

Geographic hotspot identification enables evidence-based resource pre-positioning. FEMA regional offices and supply caches should align with observed patterns:

- Southern Region: Hurricane response equipment, flood infrastructure
- Western Region: Wildfire suppression resources, evacuation capabilities

- Midwest: Tornado warning systems, shelters

Preparedness Funding

Federal grant programs (Hazard Mitigation Grant Program, Pre-Disaster Mitigation) should weight allocations by demonstrated vulnerability. Per-capita analysis ensures small states receive proportional support.

Building Codes & Zoning

Hotspot states should strengthen hazard-specific building codes: hurricane-resistant construction (Florida), wildfire-defensible space requirements (California), tornado-safe rooms (Oklahoma).

9.2 Research Directions

Fine-Scale Analysis

County-level or census tract-level analysis would reveal localized hotspots within states. Computational cartography advances enable high-resolution spatial analysis previously intractable.

Causal Analysis

Integrate geographic patterns with climate data, infrastructure age, socioeconomic factors to identify causal mechanisms. Structural equation modeling or geographically weighted regression could quantify relationships.

Predictive Modeling

Machine learning models (random forests, neural networks) trained on spatial patterns could predict future hotspot emergence. Climate change scenarios inform long-term projections.

9.3 Climate Change Context

Observed geographic patterns reflect current climate conditions. Climate change is altering disaster geography:

- Hurricane Expansion: Warmer ocean temperatures expanding hurricane zone northward
- Fire Migration: Wildfire risk spreading from traditional western hotspots
- Precipitation Shifts: Flood patterns changing with altered precipitation regimes

Geographic patterns identified here represent baseline for monitoring climate-driven shifts. Periodic re-analysis will track evolving risk landscapes.

9.4 Limitations & Caveats

This analysis has several limitations:

- Declaration Bias: Federal declarations differ from raw disaster occurrence
- Temporal Scope: 1953-2017 data may not reflect current (2024+) patterns
- State Aggregation: Masks within-state geographic variation
- Causality: Patterns shown are correlational, not causal
- Data Quality: Historical records may contain errors or inconsistencies

Despite limitations, strong concordance across multiple visualization approaches provides robust evidence for identified patterns.

10. Conclusion

This geographic analysis of US natural disaster declarations reveals clear spatial patterns in disaster distribution, with significant implications for emergency management and policy.

10.1 Key Findings Summary

Geographic Concentration

Southern states, particularly Texas, Florida, and Louisiana, dominate disaster declarations. This concentration reflects hurricane exposure, geographic extent, and climate patterns. The Gulf Coast forms a continuous high-risk corridor requiring sustained federal support infrastructure.

Type-Specific Patterns

Different disaster types exhibit distinct geographic signatures: hurricanes exclusively coastal, wildfires predominantly western, tornadoes concentrated in plains states. These patterns enable targeted, type-specific preparedness strategies.

Regional Hierarchies

South region experiences highest overall disaster frequency, followed by West, Midwest, and Northeast. However, per-capita normalization reveals different vulnerability patterns, with some smaller states showing high relative disaster burdens.

10.2 Methodological Contributions

This analysis demonstrates the value of diverse visualization approaches for geographic data:

- Multiple Perspectives: Different chart types reveal complementary insights
- Static + Interactive: Combines presentation quality with exploratory capability
- Cross-Validation: Pattern consistency across visualizations increases confidence
- Hierarchical Views: Multi-scale analysis from national to state level

10.3 Practical Applications

Results directly inform operational decisions:

- Emergency managers can pre-position resources based on hotspot geography
- Policymakers can allocate mitigation funding proportional to demonstrated vulnerability
- Planners can strengthen building codes in hazard-prone regions
- Researchers can prioritize regions for detailed local-scale studies

10.4 Future Outlook

Climate change is likely altering the geographic patterns documented here. Warming temperatures may expand hurricane zones northward, intensify western drought and wildfire conditions, and shift precipitation patterns affecting flood geography. Periodic re-analysis will track these evolving patterns.

Integration of climate model projections with observed patterns could enable predictive hotspot mapping, identifying regions likely to face increased disaster risk in coming decades. Such forecasts would enable proactive adaptation rather than reactive response.